Lecture 21

Thursday, 30 March 2023 1:27 PM

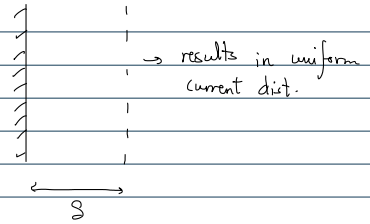
Mass Transfer effects on current distributions

↳ Mass transfer limited current distribution : + Tertiary current dist. $\nabla c \neq 0$, $t_i \to 0$ (neglect $\nabla\phi$)
(also called "convective transport" problems)

Previously, $i_{lim} \Rightarrow c_{lim} \Rightarrow 0$ @ electrode. ; constant BL thickness (δ)

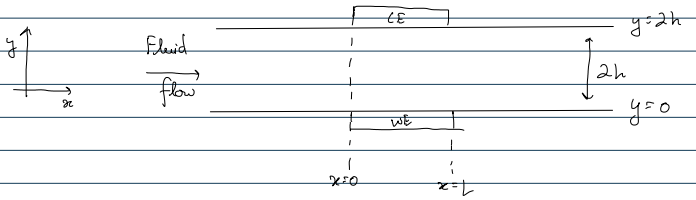
Flux is not same as conc.!

Conc = 0 @ electrode
[But there will be flux]



But geometry + flow conditions of the cell may not always result in constant δ need to
↳ we understand how M-T impacts this + current dist.

Ex: Microfluidic echem cell, w/ 2 line electrodes



in z dir, electrodes take up full width and w >> h

+ Assume laminar flow, we only have x-velocity

$$v_x(y) = \frac{3}{2} U \left(1 - \left(\frac{h-y}{h}\right)^2\right) \quad ; \quad U = \frac{Q}{2hw}$$

↳ Governing eq for $t_j = 0$: convection-diffusion (mig. neglected)

$$v_x \left(\frac{\partial c_j}{\partial x}\right) = D \left(\frac{\partial^2 c_j}{\partial x^2} + \frac{\partial^2 c_j}{\partial y^2}\right)$$

For fast kinetics: $c_{lim} = 0$ @ all $y = 0$ on electrode

Flow in the x-direction is much greater than diff. in x-direction ⇒ Microfluidic device (small Re) (convection dominates in flow direction, according to Pe)
↓
large Sc

$$Pe = \frac{\text{convective transport}}{\text{diffusive transport}} = Sc\, Re$$

Pe → large [Convection dominates]

$$Sc = \frac{\nu}{D_{lim}} = \sim 10^3 \text{ for small molecules in water}$$

$$\therefore \quad v_x \frac{\partial c_{lim}}{\partial x} = D_{lim} \frac{\partial^2 c_{lim}}{\partial y^2} \quad ; \quad \text{BC's: } \begin{array}{l} @\ x = 0,\ c_{lim} = c_\infty \quad \text{(no rxn yet)} \\ @\ y \to \infty,\ c_{lim} = c_\infty \end{array}$$

Conc. should only vary w/ δ near electrode, so we can approx velocity there w/ Taylor expansion

$$v_x = \cancel{v_x}\Big|_{y=0} + \frac{\partial v_x}{\partial y}\Big|_{y=0} y$$

(No slip)

Plugging in laminar flow eq:

$$\frac{\partial v}{\partial y} = \frac{3U}{2}\left(-2\left(\frac{h-y}{h}\right)\left(-\frac{1}{h}\right)\right)$$

$$@\ y=0 \quad \frac{\partial v_x}{\partial y} = \frac{3U}{h} \Rightarrow v_x = \frac{3Uy}{h}$$

$$\boxed{\frac{3U}{h} y \frac{\partial c}{\partial x} = D \frac{\partial^2 c}{\partial y^2}}$$

This is a PDE that can be solved w/ "comb. of variables" (Similarity solution!)

Let $\xi = y\left(\frac{U}{3hD_x}\right)^{1/3}$ → this will make the equation w/ just ind. var.

↳ make them an ODE

$$\frac{\partial c}{\partial x} = \frac{\partial c}{\partial \xi}\cdot\frac{\partial \xi}{\partial x} \quad ; \quad \frac{\partial \xi}{\partial x} = y\left(\frac{U}{3hD}\right)^{1/3}\cdot\left(-\frac{1}{3}\right)x^{-4/3} = -\frac{\xi}{3x} \Rightarrow \frac{\partial c}{\partial x} = \frac{\partial c}{\partial \xi}\cdot\left(-\frac{\xi}{3x}\right)$$

$$\frac{\partial c}{\partial y} = \frac{\partial c}{\partial \xi}\cdot\frac{\partial \xi}{\partial y} \qquad \frac{\partial \xi}{\partial y} = \left(\frac{U}{3hD_x}\right)^{1/3} = \frac{\xi}{y}$$

$\frac{\partial^2 c}{\partial y^2}$ ← Both product & chain rule

$$= \frac{\partial}{\partial y}\left(\frac{\partial c}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial c}{\partial \xi}\frac{\partial \xi}{\partial y}\right)$$

$$= \frac{\partial c}{\partial \xi}\frac{\partial}{\partial y}\left(\frac{\partial \xi}{\partial y}\right) + \frac{\partial \xi}{\partial y}\frac{\partial}{\partial y}\left(\frac{\partial c}{\partial \xi}\right)$$

$$= \frac{\partial \xi}{\partial y}\left(\frac{\partial}{\partial \xi}\left(\frac{\partial c}{\partial \xi}\right)\frac{\partial \xi}{\partial y}\right)$$

$$\frac{\partial^2 c}{\partial y^2} = \frac{\partial^2 c}{\partial \xi^2}\cdot\left(\frac{\partial \xi}{\partial y}\right)^2 = \frac{\partial^2 c}{\partial \xi^2}\cdot\left(\frac{U}{3hD_x}\right)^{2/3}$$

Plugging into original PDE:

$$\frac{3U}{h} y \frac{\partial c}{\partial \xi}\left(\frac{-\xi}{3x}\right) = D\frac{\partial^2 c}{\partial \xi^2}\left(\frac{U}{3hD_x}\right)^{2/3} \div \left(\frac{U}{3hD_x}\right)^{2/3}$$

$$3y\left(\frac{U}{3hD_x}\right)^{1/3}\frac{\partial c}{\partial \xi}(-\xi) = \frac{\partial^2 c}{\partial \xi^2}$$

ξ

$$-3\xi^2\frac{\partial c}{\partial \xi} = \frac{\partial^2 c}{\partial \xi^2} \Rightarrow \boxed{\frac{d^2c}{d\xi^2} + 3\xi^2\frac{dc}{d\xi} = 0}$$

BC's: @ $x = 0$, $c_{lim} = c_\infty \Rightarrow \xi \to \infty$ $\quad c_{lim} = c_\infty$

@ $y = \infty$, $c_{lim} = c_\infty \Rightarrow \xi \to \infty$ $\quad c_{lim} = c_\infty$

@ $y = 0$, $c_{lim} = 0 \Rightarrow @\xi = 0$, $c_{lim} = 0$

Solving using Forbenius method

$$\boxed{\frac{c_{lim}}{c_\infty} = \frac{1}{\Gamma\left(\frac{1}{3}\right)}\int_0^{\xi} e^{-x^3}\,dx}$$

→ Now we know conc. profile in BL

Solving for flux: $N_y = -D\frac{\partial c}{\partial y} = D\frac{\partial c}{\partial y}\Big|_{y=0} = D\frac{\partial c}{\partial \xi}\Big|_{\xi=0}\frac{\partial \xi}{\partial y}\Big|_{y=0}$

(Laminar flow
(No convection in y))

$$\frac{\partial}{\partial x_u}\int_0^{x_u} f(x)\,dx = f(x_u)$$

$$N_y = \frac{D c_\infty}{\Gamma\left(\frac{1}{3}\right)}\underbrace{\frac{\partial}{\partial \xi}\left(\int_0^{\xi} e^{-x^3}\,dx\right)\Big|_{\xi=0}}\left(\frac{U}{3kDx}\right)^{1/3}$$

Fundamental theorem of Calculus

$$N_y = \frac{D c_\infty}{\Gamma\left(\frac{1}{3}\right)} e^{-\xi^3}\Big|_{\xi=0}$$

$$N_y = -\frac{c_\infty D}{\Gamma\left(\frac{1}{3}\right)}\left(\frac{U}{3kDx}\right)^{1/3}$$

↙ Find current

$$\frac{i}{nF} = -N_y$$

Plot $\dfrac{i(x)}{i_{avg}}$; $i_{avg} = \dfrac{1}{L}\displaystyle\int_0^L i_{avg}\,dx$